

Cory McCartan

Curriculum Vitae

August 2026

CONTACT INFORMATION	Department of Statistics, Penn State University 325 Thomas Building, 461 Pollock Road University Park, PA 16802	(425) 770-9244 mccartan@psu.edu
ACADEMIC EMPLOYMENT	Pennsylvania State University Hoben and Patricia Thomas and Thomas and Ann Hettmansperger Early Career Professor of Statistics Assistant Professor of Statistics Affiliate Faculty in Political Science New York University Center for Data Science Data Science Assistant Professor / Faculty Fellow	2024 – 2024 – 2027 2024 – 2023 – 2024
EDUCATION	Harvard University Ph.D., Statistics, 2023. Committee: Kosuke Imai (chair), Xiao-Li Meng, Gary King. Dissertation: <i>Computational and Bayesian Methods for Geographic Data in the Social Sciences</i> . A.M., Statistics, 2021. Grinnell College B.A., Mathematics, with honors, 2019.	2019 – 2023 2015 – 2019
PEER-REVIEWED PUBLICATIONS	“Individual and Differential Harm in Redistricting,” with Christopher T. Kenny (2026). <i>Nature Human Behaviour</i> , Online ahead of print. “Redistricting Reforms Reduce Gerrymandering by Constraining Partisan Actors,” with Christopher T. Kenny, Tyler Simko, Emma Ebowe, Michael Y. Zhao, and Kosuke Imai (2026). <i>American Political Science Review</i> , Online ahead of print. “Estimating Racial Disparities when Race is Not Observed,” with Robin Fisher, Jacob Goldin, Daniel E. Ho, and Kosuke Imai (2025). <i>Journal of the American Statistical Association</i> 120:552, 2140–2153. “Evaluating Bias and Noise Induced by the U.S. Census Bureau’s Privacy Protection Methods,” with Christopher T. Kenny, Tyler Simko, Shiro Kuriwaki, and Kosuke Imai (2024). <i>Science Advances</i> 10:18, eadl2524. “Measuring and Modeling Neighborhoods,” with Jacob R. Brown and Kosuke Imai (2024). <i>American Political Science Review</i> 118:4, 1966–1985. “Letter: Census Officials Must Constructively Engage with Independent Evaluations,” with Christopher T. Kenny, Tyler Simko, and Kosuke Imai (2024). <i>Proceedings of the National Academy of</i>	

Sciences 121:11, e2321196121.

Letter to the editor re: Jarmin et al. (2023).

“Making Differential Privacy Work for Census Data Users,” with Tyler Simko and Kosuke Imai (2023). *Harvard Data Science Review* 5:4.

With response and rejoinder.

“Sequential Monte Carlo for Sampling Balanced and Compact Redistricting Plans,” with Kosuke Imai (2023). *Annals of Applied Statistics* 17:4, 3300–3323.

AoAS invited JSM session, 2024.

Covered by *The Washington Post*, *Quanta* magazine.

“Widespread Partisan Gerrymandering Mostly Cancels Nationally, but Reduces Electoral Competition,” with Christopher T. Kenny, Tyler Simko, Shiro Kuriwaki, and Kosuke Imai (2023). *Proceedings of the National Academy of Sciences* 120:25, e2217322120.

“Letter: Researchers need better access to U.S. Census data,” with Tyler Simko and Kosuke Imai (2023). *Science* 380:6648, 902–903.

“Recalibration of Predicted Probabilities Using the “Logit Shift”: Why Does It Work, and When Can It Be Expected to Work Well?” with Evan T.R. Rosenman and Santiago Olivella (2023). *Political Analysis* 31:4, 651–661.

“Comment: The Essential Role of Policy Evaluation for the 2020 Census Disclosure Avoidance System,” with Christopher T. Kenny, Shiro Kuriwaki, Evan T.R. Rosenman, Tyler Simko, and Kosuke Imai (2023). *Harvard Data Science Review*, Special Issue 2.

Response to boyd and Sarathy (2022).

“Simulated Redistricting Plans for the Analysis and Evaluation of Redistricting in the United States,” with Christopher T. Kenny, Tyler Simko, George Garcia III, Kevin Wang, Melissa Wu, Shiro Kuriwaki, and Kosuke Imai (2022). *Nature: Scientific Data* 9:1, 689.

Covered by *The New York Times* in 2025 and 2026.

“The Use of Differential Privacy for Census Data and its Impact on Redistricting: The Case of the 2020 U.S. Census,” with Christopher T. Kenny, Shiro Kuriwaki, Evan T.R. Rosenman, Tyler Simko, and Kosuke Imai (2021). *Science Advances* 7:41, eabk3283.

Originally a Public Comment to the Census Bureau (May 28, 2021).

Covered by *The Washington Post*, the *Associated Press*, the *San Francisco Chronicle*, *NC Policy Watch*, and others.

“Geodesic Interpolation on Sierpinski Gaskets,” with Caitlin Davis, Laura LeGare, and Luke Rogers (2021). *Journal of Fractal Geometry* 8:2, 117–152.

WORKING PAPERS

“Seeing the Forest for the Trees: The Gaussian Process Limit of BART,” with Melody Huang (2026). Under Review.

“Identification and Semiparametric Estimation of Conditional Means from Aggregate Data,” with Shiro Kuriwaki (2025). Under Review.

“Generalized Sequential Monte Carlo Sampling for Redistricting Simulation,” with Philip O’Sullivan and Kosuke Imai (2026). Under Review.

“Relative Bias Under Imperfect Identification in Observational Causal Inference,” with Melody Huang (2025). Under Review.

“The Role of Confounders and Linearity in Ecological Inference: A Reassessment,” with Shiro Kuriwaki (2025). Under Review.

“Gerrymandering and Geographic Polarization Have Reduced Electoral Competition,” with Ethan Jasny, Christopher T. Kenny, Tyler Simko, Melissa Wu, Michael Y. Zhao, Aneetej Arora, Emma Ebowe, Philip O’Sullivan, Taran Samarth, and Kosuke Imai (2025).

“Projective Averages for Summarizing Redistricting Ensembles” (2024).

“Finding Pareto Efficient Redistricting Plans with Short Bursts” (2024).

OTHER WRITING

“Review: Introduction to Political Analysis in R (H. Whitt Kilburn).” *The American Statistician* (2026).

“Candy Cane Shortages and the Importance of Variation.” International Statistical Institute: *Statisticians React to the News* (December 21, 2021).

“Where Will the Rocket Land?” International Statistical Institute: *Statisticians React to the News* (May 12, 2021).

“Who’s the Most Electable Democrat? It Might be Warren or Buttigieg, Not Biden.” *The Washington Post* (October 23, 2019).

“I-405 Express Toll Lanes: Usage, Benefits, and Equity,” with Shirley Leung, C.J. Robinson, Kiana Roshan Zamir, Vaughn Iverson, and Mark Hallenbeck. Technical report for the Washington State Department of Transportation (2019).

SOFTWARE

redist: Simulation Methods for Legislative Redistricting

seine: Semiparametric Ecological Inference

birdie: Bayesian Instrumental Regression for Disparity Estimation

bases: Basis Expansions for Regression Modeling

easycensus: Quickly Find, Extract, and Marginalize U.S. Census Tables

redistmetrics: Redistricting Metrics

adjustr: Stan Model Adjustments and Sensitivity Analyses using Importance Sampling

PL94171: Tabulate P.L. 94-171 Redistricting Data Summary Files

conformalbays: Jackknife(+) Predictive Intervals for Bayesian Models

alarmdata: Download, Merge, and Process Redistricting Data

blockpop: Estimate Census Block Populations for 2020

ggredist: Scales, Geometries, and Extensions of ggplot2 for Election Mapping

tinytiger: Lightweight Interface to TIGER/Line Shapefiles

adj: Lightweight Adjacency Lists

schmear: Build Structured Data Frame Subtypes

causaltbl: Tidy Causal Data Frames and Tools

onnxr: Bindings to ONNX Runtime

wacolors: Colorblind-Friendly Palettes from Washington State

nbhdmodel: Neighborhood Modeling and Analysis

PRESENTATIONS

Invited

Chalmers University of Technology. <i>Principled Statistics in the Age of AI Workshop.</i>	June 2026
Harvard Law School, Charles Hamilton Houston Institute for Race and Justice. <i>Racial Equality after Callais Convening.</i>	April 2026
Meta, Inc., Central Applied Science team. <i>Computational Social Science Seminar.</i>	April 2026
Penn State University, Clinical and Translational Sciences Institute. <i>Biostatistics, Epidemiology and Research Design (BERD) Recent Topics in Research Methods.</i>	April 2026
Sant'Anna School of Advanced Studies. <i>L'EMbeDS Data Science Seminar Series.</i>	January 2026
Oxford University, Dept. of Statistics. <i>Causal Inference Reading Group.</i>	May 2025
McMaster University, Faculty of Social Science. <i>Spark Talk Speaker Series.</i>	February 2025
University of North Carolina at Chapel Hill, Dept. of Political Science. <i>Methods and Design Workshop.</i>	February 2025
University of Wisconsin – Madison, Depts. of Political Science and Statistics. <i>Methods, Experiments, and Design Workshop.</i>	January 2025
Harvard Law School, Charles Hamilton Houston Institute for Race and Justice. <i>Guinier Project Research Roundtable.</i>	January 2025
Princeton University. <i>Frontiers in Data Science Symposium: Advances in Record Linkage.</i>	October 2024
Penn State University, Dept. of Political Science. <i>Colloquium Series.</i>	September 2024
New York University. <i>Math and Democracy Seminar.</i>	May 2024
Massachusetts Institute of Technology, Dept. of Political Science. <i>Political Methodology Speaker Series.</i>	September 2023
Harvard University, IQSS. <i>Applied Statistics Workshop.</i>	April 2023
Harvard University, IQSS. <i>Applied Statistics Workshop.</i>	October 2022
Harvard University, IQSS. <i>Applied Statistics Workshop.</i>	September 2021
Harvard University, IQSS. <i>Applied Statistics Workshop.</i>	September 2020

Conferences

Joint Statistical Meetings: 2026, 2024 (Invited paper panel), 2022 (Invited paper panel), 2021 (Invited paper panel)

Society for Political Methodology: 2026, 2025, 2024, 2023, 2022, 2022 (Poster), 2021 (Poster)

European Political Science Society: 2026

American Causal Inference Conference: 2026 (Poster), 2025, 2024 (Poster)
 Southern Political Science Association: 2026, 2025
 Small Area Estimation Conference: 2025 (Invited panel)
 Keystone State Statistics Symposium: 2024
 ACM Conference in Equity and Access in Algorithms, Mechanisms, and Optimization: 2023
 American Association for Public Opinion Research: 2022 (Poster)

TEACHING

Penn State University

STAT 497: Machine Learning and Complex Data (special topic)	Fall 2026
STAT 440: Computational Statistics	Fall 2024, Fall 2025
STAT 597: Visualization and Communication in Statistics (special topic short course)	Fall 2025
STAT 597: Missing Data (special topic short course)	Fall 2024

New York University

DS-UA 111: Data Science for Everyone	Spring 2024
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Harvard University

STAT 117: Introduction to Biostatistics (Teaching Fellow)	Spring 2021
STAT 221: Monte Carlo Methods & Other Computational Tools for Statistical Learning (Teaching Fellow)	Fall 2020

Grinnell College

MAT 215: Linear Algebra (Peer Mentor)	Fall 2017, Spring 2019
MAT 310: Statistical Modeling (Peer Mentor)	Fall 2018
Grinnell College Math Lab	2018 – 2019

ADVISING

Ph.D. Committees

Ross Cohen-Kristiansen (<i>co-chair</i> , Statistics, Penn State University)	2025 –
Tinghua Chen (Informatics, Penn State University)	2025 –
Kihyun Han (Statistics, Penn State University)	2025 –
Yang Yang (Political Science, Penn State University)	2025 –
Kyle McGovern (Bioinformatics and Genomics, Penn State University)	2025

M.S. Theses

Giovanni Stivella (Economics, University of Pisa and the Scuola Sant'Anna, Italy)	2025
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HONORS AND AWARDS

Electoral Reform and Proportional Representation grant, Protect Democracy, 2026 (total award: \$3,000).
Hoben and Patricia Thomas and Thomas and Ann Hettmansperger Early Career Professorship in Statistics, 2024 (total award: \$75,000).
SLMath Postdoctoral Fellowship (formerly MSRI), Berkeley, CA, 2023 (declined).
Best Statistical Software Award, for developing statistical software that makes a significant research contribution; awarded to the redist software package by the Society for Political Method-

ology, 2022.

Certificate of Distinction in Teaching, awarded on the basis of student feedback by the Derek Bok Center for Teaching and Learning, 2021.

Pamela Ferguson Endowed Prize, awarded to up to two senior students by the Grinnell College Department of Mathematics, 2018.

SERVICE Reviewer: *Proceedings of the National Academy of Sciences*, *Journal of the American Statistical Association*, *Journal of the Royal Statistical Society: Series A*, *Annals of Applied Statistics*, *Political Analysis*, *American Journal of Political Science*, *Quarterly Journal of Political Science*, *Harvard Data Science Review*, *ACM/IMS Journal of Data Science*, *La Matematica*, *Public Choice*, *Multi-scale Modeling and Simulation*, *Discrete Applied Mathematics*, *Election Law Journal*, *Proceedings of the IASSL Triennial International Statistics Conference*, *Sloan Foundation*.

Discussant: 2026 PolMeth Conference, 2026 SPSA Annual Conference, 2024 PolMeth Conference, 2024 MPSA Annual Conference.

Penn State University

Climate and Diversity committee, College of Science	2026 – 2027
Rao Prize committee	2026
Named lectures committee	2026
Ph.D. admissions committee	2025 – 2027
SMAC Talk committee	2025 – 2027
Colloquium committee	2024 – 2026

New York University

Faculty fellow hiring review	2023 – 2024
MA admissions committee	2023 – 2024

Harvard University

Harvard Statistics Graduate Council	2020 – 2023
Organized Ph.D. student retreat and research “lightning talks,” 2020 and 2021.	
First-year Ph.D. Student Mentor	2020 – 2023
Harvard Graduate Students Union – UAW Local 5118	2019 – 2021
Elected member, Bargaining Committee, 2020–2021 and 2021–2024 contracts.	
Interim chair, Finance and Benefits Committee, 2020.	

OTHER EXPERIENCE

CBS News	2024 –
Election consulting and election night Data Desk	
Protect Democracy	2024
Consultant, electoral reform modeling	
Data for Progress	2022
Consultant, midterm election modeling	
University of Washington eScience Institute	Summer 2019
Data Science for Social Good Fellow	
Union of Grinnell Student Dining Workers	2016 – 2019
Founder, President (2016–17), and Advisor to the Executive Board (2018–19)	

University of Connecticut REU Participant, Department of Mathematics	Summer 2018
Fred Hutchinson Cancer Research Center Lead Intern, Department of Biostatistics	Summer 2017
Cray, Inc. (now HPE) Intern, Chapel language testing	Summer 2015

EXPERT REPORTS	<i>Cubanos Pa'lante v. Florida House of Representatives</i> (U.S. District Court for the Southern District of Florida, Case No. 1:24-cv-21983). Testified by deposition and at trial. <i>Hodges v. Albritton</i> (U.S. District Court for the Middle District of Florida, Case No. 8:24-cv-879). Testified by deposition and at trial. <i>McClure et al. v. Jefferson County Commission</i> (U.S. District Court for the Northern District of Alabama, Case No. 2:23-cv-00443). Testified by deposition and at trial. <i>Callais et al. v. Landry</i> (U.S. District Court for the Western District of Louisiana, Case No. 3:24-cv-00122). Testified by deposition and at trial. <i>GRACE, Inc. et al. v. City of Miami</i> (U.S. District Court for the Southern District of Florida, Case No. 1:22-cv-24066). Testified by deposition and at trial. <i>Nairne et al. v. Ardoin</i> (U.S. District Court for the Middle District of Louisiana, Case No. 3:22-cv-0017). Testified by deposition and at trial. <i>Blair v. Department of Homeland Security</i> (U.S. District Court for the Central District of California, Case No. 2:26-cv-00417). <i>USF College Republicans v. Lutnick</i> (U.S. District Court for the Middle District of Florida, Case No. 8:25-cv-02486). <i>League of Women Voters v. Ohio Redistricting Commission</i> (Ohio Supreme Court, Case Nos. 2021-1193 and 2021-1449). Consultant (with Prof. Kosuke Imai).
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